MILITARY SPECIFICATION SHEET

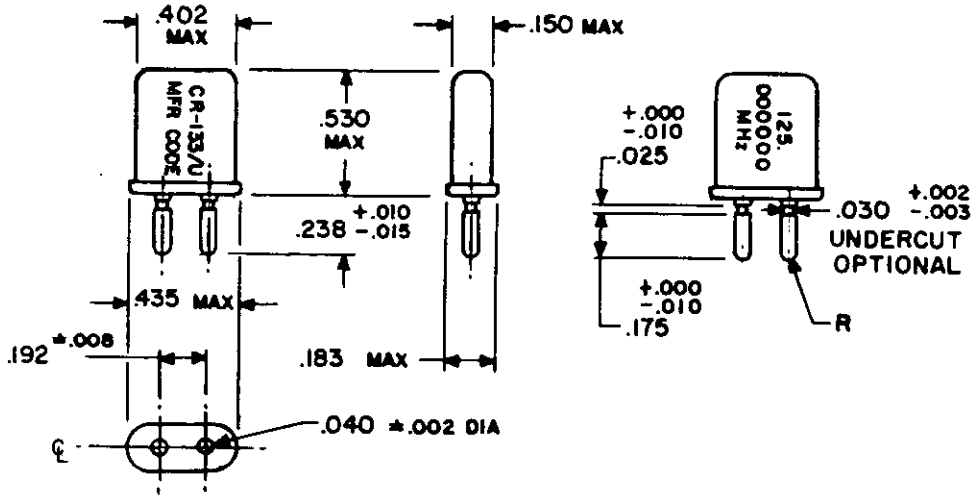
CRYSTAL UNIT, QUARTZ, CR-133/U

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(B) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 45 to 125 MHz; fifth mechanical overtone; non-controlled; series resonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.015	.38	.150	3.81	.238	6.05
.003	.08	.025	.64	.175	4.45	.402	10.21
.008	.20	.030	.76	.183	4.65	.435	11.05
.010	.25	.040	1.02	.192	4.88	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivaletns are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Crystal unit - CR-133/U.

(B) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-25/U (MIL-H-10056/11)
Dimensions, marking, and configuration	See figure 1
Frequency range	45 to 125 MHz, inclusive
Resonance	Series
Mode of oscillation	Fifth mechanical overtone
Unwanted modes	Minimum unwanted mode effective
Method II	resistance of 80 ohms or an unwanted-to-main mode resistance ratio of 2:1, whichever is greater
Operating temperature range	-55°C to +150°C, inclusive
Frequency tolerance	±0.006 percent
Equivalent resistance	60 ohms, maximum 20 ohms, minimum
Government reference standard test set	TS-683/TSM Up to 110 MHz AN/TSM-15 Above 110 MHz
Rated drive level	2.0 ±0.4 mW
Calibration values	
Resistance	60 ohms
Voltage drop	0.35 volt
Capacitance, shunt	6.5 ±1.0 pF
Shock:	
Frequency change permitted	±3 ppm
Equivalent resistance change permitted	±15 percent
Thermal shock 1/	
Frequency change permitted	±3 ppm
Equivalent resistance change permitted	±15 percent
Vibration: 2/	
Frequency change permitted	±3 ppm
Equivalent resistance change permitted	±15 percent
Aging:	
Frequency change permitted	3 ppm

1/ Method 102 of MIL-STD-202, test condition C, step 3 shall be 150°C.

2/ Method 204 of MIL-STD-202, test condition A.

B QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTES:

1. These crystal units are for use in the RCA AN/ARR-69. They were previously fabricated to the requirements specified in Air force Technical Order 12-1-20, Supplement No. 37 or RCA Drawing No. 8537694.

2. Formula for computing the crystal frequency is: $F_X = \frac{F_O - 65.0 \text{ MHz}}{2}$

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Ⓑ Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17, 19

Preparing activity:
Air Force - 85

(Project 5955-F186-4)

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